

Adult Census Income Prediction

HIGH LEVEL DOCUMENT

AGENDA

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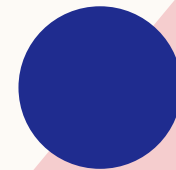
Proposed solution

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1.INTRODUCTION: WHY THIS HIGH LEVEL DOCUMENT?

THE PURPOSE OF THIS HIGH-LEVEL DESIGN (HLD) DOCUMENT IS TO ADD NECESSARY DETAIL TO THE CURRENT PROJECT DESCRIPTION TO REPRESENT A SUITABLE MODEL FOR CODING. THIS DOCUMENT IS ALSO INTENDED TO HELP DETECT CONTRADICTIONS PRIOR TO CODING, AND CAN BE USED AS A REFERENCE MANUAL FOR HOW THE MODULES INTERACT AT A HIGH LEVEL.

THE HLD WILL:

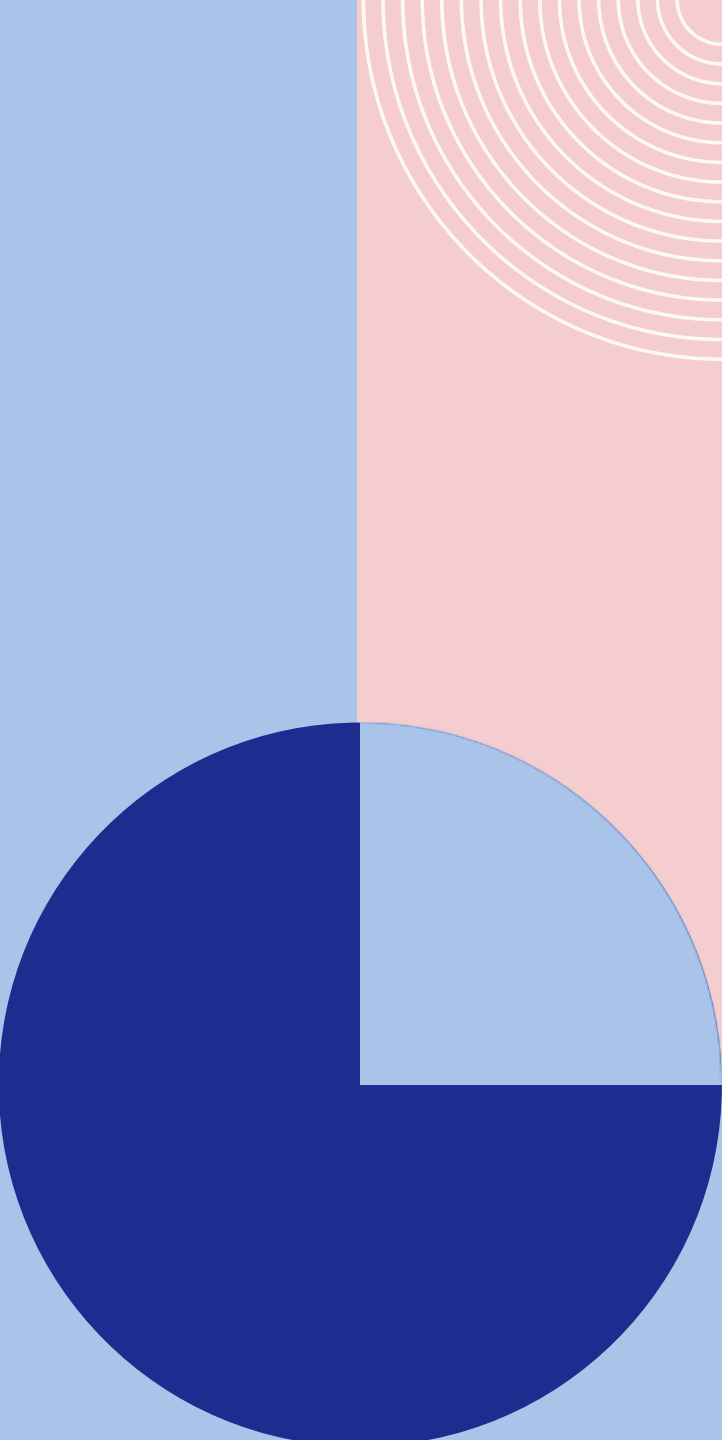
- .SHOWCASE EVERY DESIGN ELEMENT AND PROVIDE A DETAILED DEFINITION OF IT.
- .DESCRIBE EVERY USER INTERFACE THAT IS BEING USED.
- .EXPLAIN THE INTERFACES FOR THE HARDWARE AND SOFTWARE.
- .SPECIFY THE NECESSARY PERFORMANCE STANDARDS.
- .INCLUDE THE PROJECT'S ARCHITECTURE AND DESIGN ELEMENTS.
- .DESCRIBE AND ENUMERATE THE NON-FUNCTIONAL CHARACTERISTICS, SUCH AS:
 - .SAFETY
 - .DEPENDABILITY
 - .RELIABILITY
 - .MOBILITY
 - .RETURNABILITY
 - .COMPATIBILITY OF APPLICATIONS
 - .USE OF RESOURCES

Scope:

The system architecture, including the database architecture, application architecture (layers), application flow (navigation), and technology architecture, is presented in the HLD documentation. The HLD use language that is either non-technical or moderately technical, so system administrators should be able to grasp it.

GENERAL DESCRIPTION

ACIP	Adult Census Income Predictor
Database	Collection of the Information monitored by the system
Cloud	A datacenter full of servers connected to the internet performing a service
IDE	Integrated Development Environment
UI	User Interface
Anvil	A Python based UI builder
Heroku	A cloud service



PRODUCT DESCRIPTION

The ACIP is a classification model that uses machine learning to assist us anticipate an individual's income based on a set of factors.

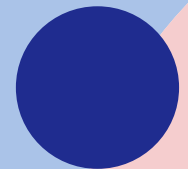
PROBLEM STATEMENT

to develop an AI-powered tool that can forecast an individual's yearly income and implement it as a user interface.

The objective is to forecast whether or not an individual makes more than \$50,000 annually. In essence, this is a binary classification problem in which an individual is assigned to either the >50K group or the ≤50K group.

PROPOSED SOLUTION

employing all of the common methods used in a data science project's lifecycle, such as data exploration, data cleaning, feature engineering, model selection, model building, and model testing, in addition to developing a front end where users can enter data and receive results instantly.

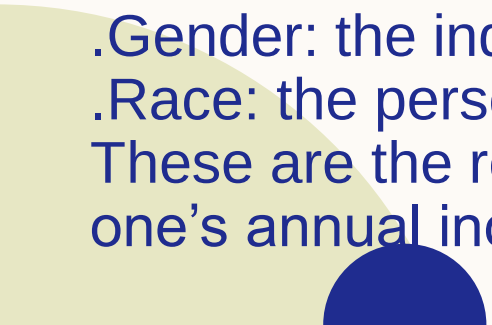
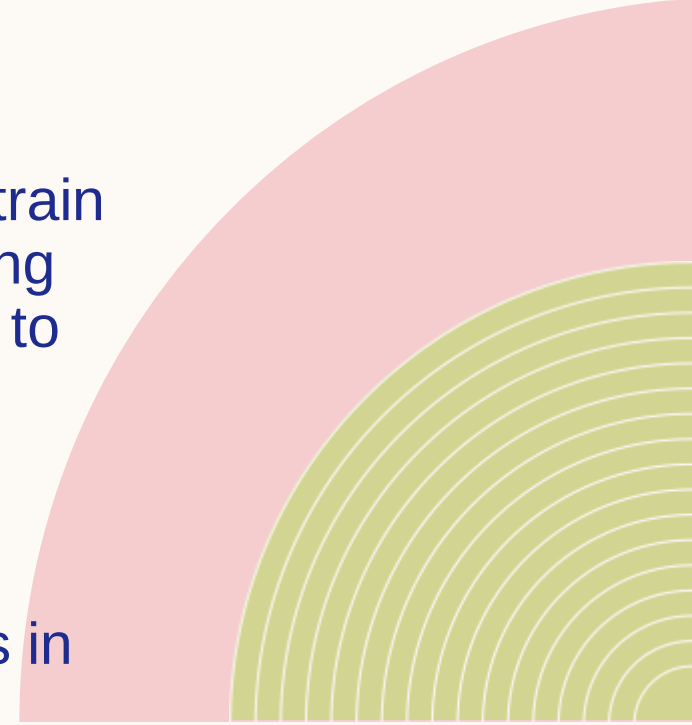


DATA REQUIREMENTS

Our problem statement is the only source of information needed. To train our model, we require the dataset from the 1994 census. The following characteristics are necessary for every record in the dataset in order to calculate an individual's income:

- .Age: The age of the individual for whom the income needs to be calculated.
- .Hours worked: The amount of time a person puts in each week.
- .Gain/Loss in Capital: The individual's total amount of gains or losses in capital, if any.
- .Country: the nation where the individual resides.
- .Education Level: the most degree of education the individual has received.
- .Marital Status: the individual's state of being married or single
- .occupation: the person's line of work.
- .Work class: The individual's working class.
- .Gender: the individual's gender.
- .Race: the person's race of origin.

These are the required set of parameters required in order to predict one's annual income

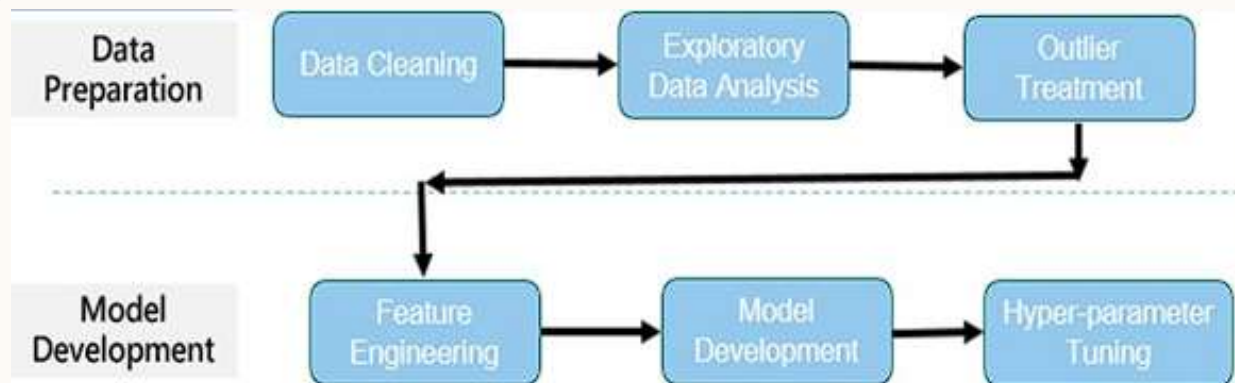


TOOLS USED

Python programming language and frameworks such as NumPy, Pandas, Scikit-learn, Flask, Anvil, and a few other libraries were used to build the whole model.

- Google colab and Visual Studio code were used as IDE
- For visualization tasks, matplotlib, seaborn and plotly
- Heroku was used for deployment of the model
- Anvil and Flask were used for building the web application and server to run the code
- Apache Cassandra was used to storage and retrieval of data
- Github is used as version control system
- NumPy and Pandas were used to clean and interpret data
- Scikit learn was used to cross validate and compare different models.
- CatBoost Classifier was used to build the final model

PROPOSED METHODOLOGY



Deployment Process

This was deployed Locally.

CONCLUSION

- The ACIP has the ability to assist several organizations, agencies, businesses, etc. worldwide with a variety of jobs and will provide an individual's income projections instantaneously.



**THANK
YOU**

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